EMOTION-AWARE AGILE SYSTEM FOR PROGRESS TRACKING, EXPERTISE RECOMMENDATION, REQUIREMENT MANAGEMENT AND INCLUSIVE COMMUNICATION

Project ID: 25-26J-525

Project Proposal Report

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor: (Ms. Ishara Weerathunga)	Date 30/8/2025.

ABSTARCT

Agile software development has become the dominant methodology for modern teams, but existing tools often overlook the emotional dynamics that directly influence productivity, collaboration, and sprint outcomes. This study aims to address that gap by developing an Emotion-Aware Agile System that integrates emotional intelligence into progress tracking, expertise recommendation, requirement management, and inclusive communication. The research problem focuses on how unrecognized emotional states—such as frustration, demotivation, or burnout—lead to inefficiencies in Agile workflows and conflict within teams.

The proposed solution adopts a multi-module AI architecture combining Natural Language Processing (NLP), facial and micro-expression recognition, semantic embeddings, and predictive analytics. The system includes an emotion detection engine to classify team members' emotions from chat and video data, an Agile optimization module that maps emotional signals to metrics such as velocity and workload distribution, an expertise recommendation system that uses developer embeddings and task similarity scoring for intelligent task assignment, a sprint replanning tracker that predicts the impact of requirement changes and generates revised backlogs, and an inclusive brainstorming assistant that rephrases and enhances hesitant inputs to ensure equal participation. Together, these modules form a proactive decision-support assistant for Scrum Masters and Product Owners.

Through this design, the project aspires to build the first integrated framework that links emotional intelligence with Agile practices in hybrid English-speaking development teams. The anticipated contribution is a novel system that not only monitors but also responds to emotional cues in real time, offering actionable recommendations for workload balancing, conflict mitigation, and inclusive communication. This research advances the intersection of affective computing and Agile software engineering, with the potential to create more resilient, collaborative, and human-centered development environments.

Keywords: *emotion detection engine, Agile optimization, expertise recommendation, sprint replanning tracker, inclusive brainstorming assistant*

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1. INTRODUCTION

Agile software development has emerged as the predominant methodology for contemporary software engineering teams, prized for its iterative nature, adaptability to change, and emphasis on collaborative dynamics [10]. While frameworks such as Scrum and Kanban provide robust structures for task management and workflow optimization, they predominantly address technical and procedural aspects, largely neglecting the critical human factors that fundamentally influence team performance and project outcomes. Research in organizational psychology and software engineering consistently demonstrates that emotional states—including frustration, demotivation, and burnout—significantly impact productivity, collaboration quality, and ultimately, sprint success [9]. Traditional Agile management tools (e.g., Jira, Trello) excel at tracking tangible metrics but remain fundamentally "emotion-blind," creating a critical gap in holistic project management

1.1. Background and Literature Survey

The theoretical foundation for this research intersects affective computing, software engineering, and organizational behavior. Emotion recognition in software engineering has gained traction with studies demonstrating that developer affect correlates with productivity metrics. Mäntylä et al. [3] established that sentiment analysis of commit messages can predict software defects, while Graziotin et al. [2] demonstrated links between developer happiness and self-assessed productivity.

Multimodal emotion recognition represents the state-of-the-art in affective computing, with research showing superior accuracy through information fusion from multiple channels. Poria et al. [4] provided a comprehensive survey of fusion techniques, noting that context-aware fusion outperforms unimodal approaches. In textual analysis, transformer-based models like BERT [5] and RoBERTa [6] have set new benchmarks in emotion classification from short texts, though their application to technical communication remains underexplored.

Facial expression analysis has evolved from basic emotion classification to sophisticated micro-expression recognition. Ekman's pioneering work [7] established micro-expressions as reliable indicators of concealed emotions. Recent advances by Li et al. [8] and Yap et al. [9] have developed computational methods for detecting these brief expressions (≤500ms) using optical flow and deep learning techniques, offering potential for identifying unconscious stress signals in workplace settings.

Temporal modeling of emotions has been implemented using recurrent architectures. Lee et al. [10] demonstrated the effectiveness of LSTMs for tracking emotion trajectories in conversations, while others have applied similar approaches to predict emotional shifts in team environments [11].

Despite these advances, the integration of these technologies into Agile workflows remains nascent. Current tools operate in isolation, lacking the contextual integration necessary for meaningful application in software development environments

1.2. Research Gap

A comprehensive analysis of current literature reveals several significant research gaps:

1. **Modality Isolation**: Existing affect detection systems typically operate unimodally, either analyzing text [3] or facial expressions [12], but rarely integrating multiple channels in a context-aware manner [4].
2. **Domain Specificity**: Current emotion recognition systems are trained on general datasets, lacking specialization for the technical communication and stress patterns characteristic of software development environments [2].
3. **Temporal Limitations**: Most systems provide snapshot affect analysis without modeling emotional trajectories across development cycles [10], despite evidence that emotional patterns evolve significantly throughout sprints [11].
4. **Integration Deficiency**: No existing system successfully integrates emotion recognition with Agile metrics (velocity, burn-down rates, defect density) to provide actionable insights for project management [1].
5. **Micro-expression Neglect**: Commercial affect recognition systems (e.g., Affectiva, Microsoft Emotion API) focus exclusively on macro-expressions, ignoring the critical information contained in micro-expressions that often reveal concealed stress or disagreement [7].
6. **Confidence Awareness**: Current systems rarely provide confidence metrics for their predictions, potentially leading to erroneous decisions based on unreliable classifications.

This research aims to address these gaps by developing a multimodal, temporally-aware emotion recognition system specifically designed for Agile software development environments, with robust integration to project management metrics.

Existing Research vs. Proposed Component

Legend: ✓ = Feature supported; ~ = Partially supported / Limited functionality; ✗ = Not supported / not primary focus

Features	**A**	**B**	**C**	**Proposed Component**
Text-Based Emotion Analysis	✗	✗	✓ (Basic Sentiment)	✓ (Advanced) Fine-grained emotions (anger, joy, etc.) from dev communication
Facial Macro-Expression Analysis	✓ (Primary Focus)	✓ (Primary Focus)	✗	✓ Real-time classification of basic emotions from video.
Facial Micro-Expression Analysis	✗	✗	✗	✓ (Novelty) Detects brief, concealed expressions of stress/tension (<0.5s).
Multimodal Fusion (Text + Visual)	✗	✗	✗	✓ (Novelty) Context-aware fusion dynamically weights modalities based on signal quality.
Temporal Emotion Modeling	~ (Session-based)	✗	✗	✓ (Novelty) Tracks emotion evolution over meetings/sprints and predicts future states.
Agile Metrics Integration	✗	✗	✗	✓ (Novelty) Links emotional trends to velocity, backlog distribution, and blocker resolution.
Emotion Confidence Scoring	~ (Internal)	~ (Internal)	✗	✓ (Novelty) Provides confidence scores for each prediction and aggregates them at the team level.
Real-Time Analysis in Meetings	✓	✓	✓	✓ Processes both video and

				text in real-time during Agile ceremonies
Proactive Alerting & Dashboards	~ (Generic Dashboards)	✗	~ (Basic Sentiment Graphs)	✓ Decision-support dashboard with alerts and insights tailored for Scrum Masters.

Table 1.1: Existing Research Comparison 1

A: Affectiva [16] | **B:** Microsoft Cognitive Services Emotion API [17] | **C:** Generic Slack Sentiment Analysis Tools (e.g., SentimentBot [18])

1.3. Research Problem

The core research problem addressed by this study is the absence of an integrated, emotionally-intelligent framework capable of detecting, interpreting, and responding to developer affective states within Agile software development environments. This problem manifests concretely through several interrelated challenges:

1. **Unrecognized Emotional Signals**: Subtle but critical emotional cues in developer communication (textual and visual) frequently go undetected by current project management tools, leading to unaddressed frustration, stress, and disengagement [2].
2. **Disconnect Between Affect and Performance**: Existing Agile tools track technical metrics without capturing their relationship to team emotional dynamics, preventing early intervention in productivity deterioration [1].
3. **Ineffective Intervention Timing**: The lack of temporal emotion modeling results in reactive rather than proactive management approaches, with interventions occurring only after problems have manifest in delayed sprints or quality issues [11].
4. **Homogeneous Participation**: Communication patterns in Agile ceremonies often privilege more vocal team members, while those experiencing hesitation or communication anxiety contribute less effectively, potentially losing valuable insights [13].
5. **Concealed Emotional States**: The absence of micro-expression analysis capabilities means that concealed stress, disagreement, or anxiety remain undetected, despite their significant impact on team dynamics and individual performance [7].

This research posits that these challenges can be addressed through a multimodal emotion recognition system that integrates textual, vocal, and visual analysis with temporal modeling and Agile metric correlation, providing a comprehensive emotional intelligence framework for software development teams.

2. OBJECTIVES

2.1. Main Objective

The primary objective of this research is to design and develop an integrated, emotion-aware AI assistant capable of enhancing both the productivity and well-being of Agile software development teams. Unlike traditional project management tools that focus solely on task tracking and performance metrics, this system seeks to bridge the gap between human emotional intelligence and process efficiency. It will achieve this by processing multimodal communication data collected from platforms such as Slack, GitHub, and Jira, in addition to real-time video and audio feeds during meetings and collaborative sessions. By analyzing these diverse sources, the AI assistant can detect subtle emotional cues, identify communication patterns, and understand team dynamics in context, offering a holistic view of team health that goes beyond conventional analytics.

Leveraging the insights derived from emotional and contextual data, the system will correlate these findings with core Agile performance indicators such as velocity, sprint completion rates, task dependencies, and quality metrics. This correlation will enable the assistant to provide proactive, data-driven recommendations for workload balancing, conflict resolution, and strategic sprint replanning. By delivering timely alerts, actionable guidance, and predictive analytics, the system aims to foster a more collaborative, efficient, and psychologically safe work environment. Ultimately, this research seeks to create a responsive, emotionally intelligent support tool that empowers Agile teams to manage human factors alongside technical objectives, ensuring sustainable performance, improved morale, and enhanced decision-making throughout the software development lifecycle.

2.2. Specific Objective

The specific objectives for the Emotion Monitoring System component are:

1. To design and implement a multimodal data acquisition pipeline that collects and preprocesses textual data from Agile communication platforms (Slack, GitHub commits/comments) and video streams from team meetings.
2. To develop and fine-tune domain-specific emotion classification models including:

3. A transformer-based text emotion classifier trained on annotated developer communications
4. A facial expression recognition system capable of detecting both macro-expressions and micro-expressions
5. A confidence-aware fusion mechanism that dynamically integrates multimodal predictions
6. To implement temporal modeling capabilities using recurrent neural networks to track emotion trajectories across meetings and development cycles, enabling prediction of emotional trends and potential risks.
7. To develop correlation algorithms that integrate emotional data with Agile performance metrics (velocity, workload distribution, participation levels) to provide contextualized insights.
8. To design and validate a dashboard interface that presents emotional insights with confidence metrics in an intuitive format for Scrum Masters and team members.
9. To evaluate the system's accuracy, reliability, and usefulness through controlled experiments and pilot deployment with software development teams.

03. METHODOLOGY

The development of the Emotion-Aware Agile Assistant will follow a systematic, iterative methodology grounded in robust software engineering and AI principles. The first phase involves the acquisition and curation of multimodal datasets, including anonymized team communications from platforms like Jira and Slack, pre-existing facial image datasets, and Agile metrics pulled via API. This heterogeneous data will be rigorously preprocessed: text will be tokenized, images will be normalized and augmented, and time-series Agile data will be structured for temporal analysis to create a solid foundation for model training.

Subsequent phases focus on core AI development and systems integration. Individual sub-modules—including a CNN for facial emotion recognition, fine-tuned transformers for text-based emotion detection, and tree-based algorithms like XGBoost for predicting requirement change impacts—will be developed and validated independently. These components will then be unified into a scalable microservices architecture, containerized with Docker and orchestrated with Kubernetes. The final system will be accessed through an intuitive React.js dashboard, providing real-time analytics and decision-support, before undergoing comprehensive unit, integration, and user acceptance testing to ensure practical viability and robustness.

3.1 System Architecture

The system diagram illustrates an Agile Intelligence Platform composed of four independent yet complementary components: Emotion Monitoring, Expertise Recommendation, Sprint Replanning Tracker, and Inclusive Brainstorming. Each component operates autonomously but contributes to a holistic view of Agile team performance. The Emotion Monitoring System integrates multimodal data such as text (from messaging tools), facial micro-expressions (from video conferencing), and voice tone (from voice capture services) to detect emotions in real time. It tracks team mood evolution over multiple meetings, assigns confidence scores to emotional insights, and links these trends with Agile metrics like sprint velocity, backlog balance, and blocker resolution time. This predictive and context-aware fusion ensures reliable emotion insights, allowing Scrum Masters to make informed decisions regarding team dynamics and workload balance.

The Expertise Recommendation System processes data from Jira, GitHub, and DevOps tools to build semantic developer profiles using embeddings, enabling accurate task-to-expert matching with ranked recommendations. Meanwhile, the Sprint Replanning Tracker monitors backlog and requirement changes, predicting their impact on velocity, schedule, quality, and productivity. It provides proactive mitigation strategies and generates revised sprint backlogs dynamically. Finally, the Inclusive Brainstorming Platform empowers team members to contribute ideas anonymously or with enhanced clarity, offering rephrasing support, diversity analytics, clustering, and summarization of inputs. Together, these independent modules ensure that Agile teams benefit from emotion-aware decision-making, expertise-driven task allocation, adaptive sprint replanning, and inclusive collaboration.

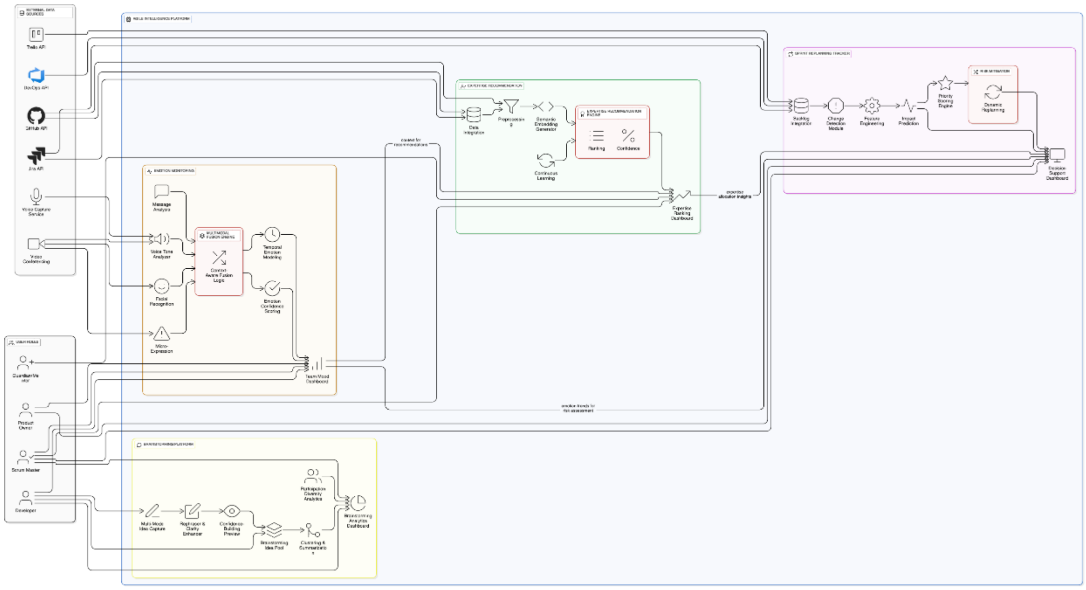

Figure 3.1: System diagram

3.2. Component Architecture

The architecture for the Multimodal Emotion Detection System is designed as an integrated pipeline that processes and fuses specific data sources relevant to team communication and meeting dynamics. The system's workflow, depicted in Figure 1, is structured into distinct layers to ensure modularity and clarity of function.

Data ingestion occurs at the Data Acquisition Layer, which is explicitly configured to gather unstructured text from Slack messages and reactions and Jira ticket descriptions and comments, alongside video streams from team meetings via webcam feeds. This focused data collection ensures the analysis is centered on direct human communication and non-verbal cues, rather than code-related metadata.

The raw data is subsequently prepared in the Preprocessing Module. Here, text from Slack and Jira undergoes cleaning (removal of URLs, special characters), tokenization, and normalization. Simultaneously, video frames are processed using computer vision libraries (OpenCV, MediaPipe) for real-time face detection, alignment, and cropping to standardize input for visual analysis.

Processed data then feeds into two parallel, specialized analysis engines:

- The Text Analysis Engine utilizes a pre-trained RoBERTa transformer model, fine-tuned on a curated dataset of workplace and developer communication, to perform emotion classification on the text from Slack and Jira, outputting categorical labels and confidence scores.

- The Visual Analysis Engine employs a dual-path approach. A Convolutional Neural Network (CNN) analyzes preprocessed frames for prototypical macro-expressions, while a separate module using optical flow and a CNN-LSTM architecture is dedicated to spotting and classifying brief micro-expressions indicative of concealed affect.

- The outputs from these engines are synthesized within the Multimodal Fusion & Analysis Core**.** A central Context-Aware Fusion Engine leverages an attention-based mechanism to dynamically weight the contribution of each modality (text and visual) based on its instantaneous signal quality and confidence score, creating a single, robust emotional assessment. This fused output is then fed into a Temporal Modeling LSTM network**,** which aggregates individual data points over time to model the emotional trajectory of each team member across meetings and sprints. This temporal context is finally enriched by integration with Agile performance metrics (e.g., sprint velocity, backlog status) fetched directly from the Jira API**,** enabling the correlation of emotional trends with project progress.
- The final processed intelligence is delivered through Decision Support outputs**:** a Scrum Master Dashboard for visualizing trends and alerts, and a proactive AI Chat Assistant integrated into Slack, providing real-time, context-aware recommendations to enhance team well-being and productivity. This end-to-end architecture ensures a closed loop from data acquisition to actionable insight, specifically tailored to improve human dynamics within Agile frameworks.

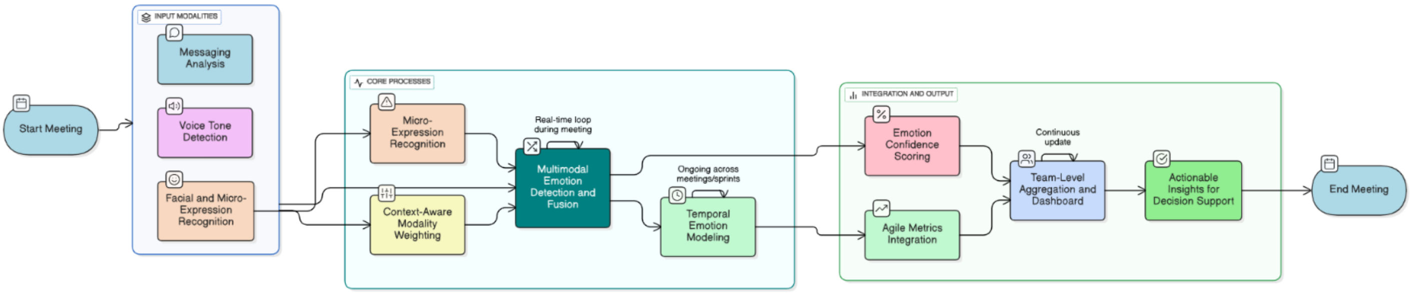

Figure 3.2: Component System Diagram

3.3 Technologies

The development of the Emotion-Aware Agile Assistant will leverage a modern, robust technology stack to handle the complex demands of multimodal data processing, machine learning, and real-time analytics.

- **Facial Emotion Recognition:** TensorFlow, Keras, PyTorch

- **Computer Vision & Facial Landmark Detection:** OpenCV, Dlib
- **Natural Language Processing (NLP):** Transformer-based models (BERT, RoBERTa), SpaCy, NLTK
- **Backend Framework:** Flask (Python), Node.js
- **Frontend Framework:** React.js
- **Database:** MongoDB (for unstructured data like logs and embeddings), PostgreSQL (for structured Agile metric data)
- **Cloud & Deployment:** AWS (Amazon Web Services) for scalable computing and storage
- **Containerization & Orchestration:** Docker, Kubernetes

04. PROJECT REQUIREMENTS

4.1 Functional Requirements

1. The system shall authenticate with and continuously collect text data from specified Slack channels and GitHub repositories
2. The system shall capture and process video streams during virtual meetings with team member consent
3. The system shall classify emotions from text inputs into seven categories (Anger, Joy, Sadness, Fear, Surprise, Neutral, Frustration) with confidence scores
4. The system shall detect and classify both macro-expressions and micro-expressions from video inputs
5. The system shall implement attention-based fusion of textual and visual emotion predictions
6. The system shall track emotional trajectories for individual team members across multiple time scales (within meeting, across sprints)
7. The system shall correlate emotional data with Agile metrics from Jira/GitHub APIs
8. The system shall generate alerts and recommendations for Scrum Masters when concerning emotional patterns are detected

9. The system shall provide visualization dashboards showing emotional trends and correlations with performance metrics

4.2 Non-Functional Requirements

1. **Performance**: Video processing pipeline shall operate with <5s latency for real-time feedback
2. **Accuracy**: Multimodal fusion shall achieve minimum 15% improvement in F1-score over best unimodal approach
3. **Reliability**: System shall maintain 99% uptime during scheduled meeting times
4. **Privacy**: All video data shall be processed locally or encrypted in transit, with no permanent storage without explicit consent
5. **Scalability**: System shall support up to 10 concurrent team meetings with 8 participants each
6. **Usability**: Dashboard shall achieve SUS score of ≥80 in user testing

4.3 System Requirements

- **Server Configuration**: 8+ vCPUs, 16GB+ RAM, NVIDIA GPU with 8GB+ VRAM (for model inference)
- **Client Requirements**: Modern browser with WebRTC support, 720p+ webcam
- **Storage**: 100GB+ SSD storage for model weights and temporary data processing
- **Network**: 100Mbps+ stable internet connection for video streaming and API communications

4.4 User Requirements

1. **Product Owner:** To understand the potential impact of a new requirement or priority change before committing it to a sprint, enabling informed trade-off decisions.
2. **Scrum Master / Team Lead:** To receive automatic alerts on disruptive changes and get data-driven replanning options to protect the team's focus and keep the sprint on track.

3. **Development Team:** To gain visibility into how changes affect their workload and potential quality risks, allowing for proactive adjustments and sustainable productivity.
4. **Stakeholders:** To receive clear, quantifiable reports on how change requests impact delivery timelines and quality, setting realistic business expectations.

05. DESCRIPTION OF PERSONAL AND FACILITIES

A detailed project plan has been formulated to ensure the systematic and timely execution of this research. The plan is structured into five sequential phases, encompassing all activities from initial planning to final thesis submission. The following Work Breakdown Structure (WBS) delineates the specific tasks, their ownership, and interdependencies. Subsequently, a Gantt chart provides a clear visual timeline for the project's 7-month duration, highlighting the critical path and parallel development efforts, particularly in the model development phase. This structured approach is designed to facilitate effective project management, ensure comprehensive coverage of all objectives, and mitigate the risk of delays

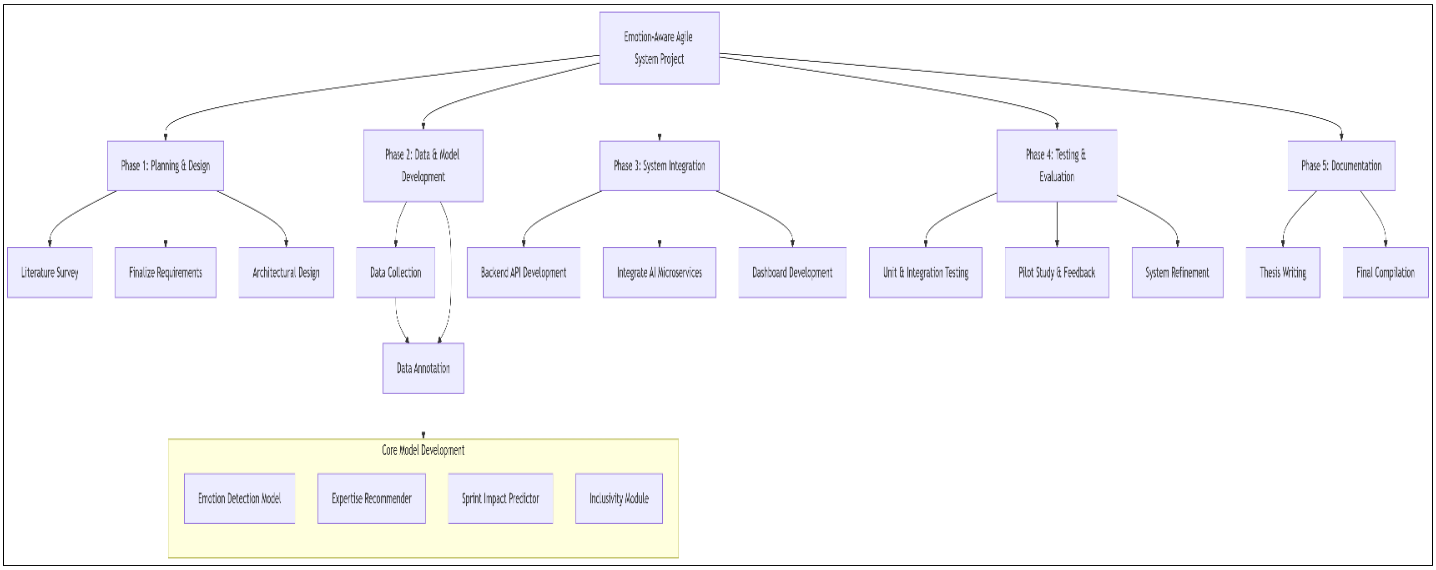

Figure 5.1: Work Breakdown Structure

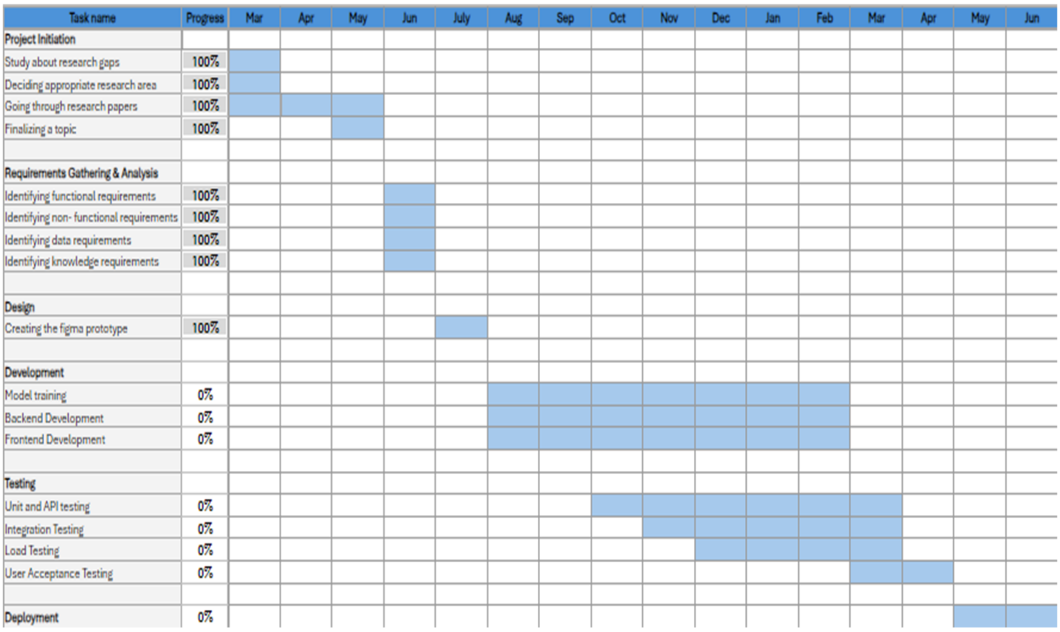

Figure 5.2: Gantt Chart

06. BUDGET AND BUDGET JUSTIFICATION

Category	**Description**	**Estimated Cost (Rs.)**
Internet Charges	Monthly broadband charges during project	3,000.00
Development	Expenses for coding tools, IDE licenses, or plugins	5,000.00
Technical Information Learning	Books, tutorials, and online learning materials	4,500.00
Online Resources & Subscriptions	Research platforms, premium accounts for learning	3,000.00
APIs	Paid APIs for integration and testing	6,000.00
Cloud Services	Hosting, compute, and storage services	12,000.00
Databases	Cloud database subscription or license fees	7,500.00

Table 6.1: Overall Budget For System

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